TASK – 4

INTELLIGENT DNA PATTERN SEARCH Java

PROGRAM :

```
import java.util.*;
import java.util.stream.*;

public class DNAPatternSearch {
   public static void main(String[] args) {

      String dna = "ATGCGATATGCG";
      String pattern = "ATG";

      List<Integer> positions =
         IntStream.rangeClosed(0, dna.length() - pattern.length())
            .filter(i -> dna.startsWith(pattern, i))
            .boxed()
            .collect(Collectors.toList());

      System.out.println("DNA Sequence: " + dna);
      System.out.println("Search Pattern: " + pattern);

      if (positions.isEmpty()) {
         System.out.println("Pattern not found.");
      } else {
```

```
        System.out.println("Pattern found at positions: " + positions);
        System.out.println("Total occurrences: " + positions.size());
    }
  }
}
```

INPUT :

DNA Sequence: ATGCGATATGCG

Search Pattern: ATG

OUTPUT :

DNA Sequence: ATGCGATATGCG

Search Pattern: ATG

Pattern found at positions: [0, 7]

Total occurrences: 2